SUPERSTORE SALES & PROFITABILITY ANALYSIS

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Project	Superstore Giant — Sales & Profitability Analysis
Tools Used	Microsoft Excel | SQL Server | Power BI
Dataset	Superstore Sales Dataset (9,994 rows, 2014–2017)
Date	June 2026

Objective:
To analyze sales performance, profitability drivers, customer behavior, and the impact of discounting on business outcomes using Excel, SQL, and Power BI.

Disclaimer:
This project was completed for portfolio and learning purposes using the publicly available Superstore dataset on Kaggle. Any company names, recommendations, or business scenarios presented in this report are fictional and intended solely to demonstrate data analytics and business intelligence skills.

1. Executive Summary

This portfolio project presents an end-to-end analysis of retail sales and profitability data spanning four years (2014–2017). The project demonstrates the application of Microsoft Excel for data cleaning, SQL Server for exploratory and business-focused analysis, and Power BI for dashboard development and data visualization.

The objective was to evaluate overall business performance, identify the drivers of profitability, assess the impact of discounting strategies, and provide data-driven recommendations for improving financial performance. The analysis revealed that while revenue and order volume increased significantly over time, excessive discounting was consistently associated with declining margins, loss-making products, and underperforming regions.

Headline Numbers

$2,297,201 Total Revenue	**$286,398** Total Profit	**$156,131** Total Loss	**12.47%** Profit Margin	**5,009** Total Orders

Key Findings at a Glance

- Revenue grew 51% from 2014 to 2017, but profit margin dipped in the highest-revenue year (2017) — signaling a scaling problem
- 18.7% of all orders (1,871 rows) are loss-making, generating a combined loss of $156,131
- 856 orders with discounts above 50% alone cost the business $76,559 in losses
- Furniture is a near-breakeven category generating only 2.49% profit margin on $742K in revenue
- 10 states are operating at a total loss, collectively destroying $97,687 in profit
- The Central region carries a 24% average discount — more than double the West region — and has the lowest profit margin at 7.92%
- Eliminating all discounts above 50% would increase total profit by 26.7% overnight

2. Data Cleaning Summary (Microsoft Excel)

The raw dataset was loaded into Microsoft Excel and subjected to a full data quality audit before any analysis was performed. The following checks were carried out systematically.

2.1 Data Quality Checks

Check Performed	Finding	Action Taken	Status
Duplicate Rows (Row ID)	None found	No action required	**PASS**
Missing / Null Values	All 9,994 rows complete across all 20 columns	No action required	**PASS**
Date Column Formats	Order Date & Ship Date correctly formatted as MM-DD-YYYY	No action required	**PASS**
Numeric Column Types	Sales, Quantity, Discount, Profit all right-aligned and numeric	No action required	**PASS**
Discount Format	Stored as decimals (0.2 = 20%)	Confirmed correct for calculations	**PASS**
Negative Shipping Days	Minimum = 0, Maximum = 7 days	No impossible dates found	**PASS**
Negative Profit Orders	1,871 rows (18.7%) with negative profit	Flagged for analysis	**FLAGGED**
High Discount Orders (>50%)	856 orders identified	Flagged for analysis	**FLAGGED**

2.2 Calculated Columns Added

Column Name	Formula	Finding
Days to Ship	= Ship Date - Order Date	Range: 0 to 7 days. No negative values. Same-day shipping exists.
Profit Margin %	= (Profit / Sales) x 100	Range: -275% to +50%. Extreme negative margins confirm deep losses on certain orders.
High Discount Flag	= IF(Discount > 0.5, 'High Discount', 'Normal')	856 High Discount orders (8.6%) vs 9,138 Normal orders.

KEY FINDING	The data cleaning phase revealed a structurally clean dataset with no duplicates or missing values. However, two critical business flags emerged: 1,871 loss-making orders totaling -$156,131 and 856 high-discount orders (>50%), both pointing to a systemic discounting problem requiring urgent attention.

3. SQL Analysis Findings

The cleaned dataset was imported into SQL and analyzed across five sections covering sales trends, category performance, regional analysis, customer segments, and shipping operations. A total of 18 queries were executed.

3.1 Sales & Profit Overview

3.1.1 Overall Business Performance

The Superstore generated $2,297,201 in total revenue and $286,398 in profit at a 12.47% overall margin. While these headline numbers appear reasonable, the margin is below the healthy retail benchmark of 20-30%, indicating significant room for improvement.

3.1.2 Yearly Revenue and Profit Trend

Year	Revenue	Profit	Margin %	Orders
2014	$484,247.56	$49,544.06	10.23%	969
2015	$470,532.46	$61,618.69	13.10%	1,038
2016	$609,205.86	$81,795.27	13.43%	1,315
2017	$733,215.19	$93,439.77	**12.74%**	1,687

- Revenue grew 51% from 2014 to 2017 — a strong top-line performance
- Order volume nearly doubled from 969 to 1,687 orders
- WARNING: Profit margin declined in 2017 (12.74%) despite being the highest revenue year — the business is growing but becoming less profitable per dollar of sales

3.1.3 Monthly Sales Seasonality

- Top 3 revenue months: November ($352K), December ($325K), September ($307K)
- Weakest months: February ($59K) and January ($94K) — a significant Q1 revenue gap
- November generates the highest revenue but lower profit than December — Black Friday discounting effect is clearly eroding margins
- Q4 (Oct-Dec) accounts for approximately 38% of annual revenue

3.1.4 Top & Bottom Products by Profit

Product	Category	Profit	Margin %
Canon imageCLASS 2200 Advanced Copier	Technology	**$25,199.94**	40.91%
Fellowes PB500 Binding Machine	Office Supplies	$7,753.06	28.24%
Hewlett Packard LaserJet 3310 Copier	Technology	$6,983.89	37.07%
Cubify CubeX 3D Printer Double Head	Technology	**($8,879.97)**	-80.00%
Lexmark MX611dhe Laser Printer	Technology	**($4,589.97)**	-27.27%
Chromcraft Bull-Nose Conference Table	Furniture	**($2,876.11)**	-29.00%

- The Canon imageCLASS 2200 Copier alone contributes ~9% of total company profit
- The Cubify CubeX 3D Printer (Double and Triple Head combined) lost $12,719 — the more units sold, the greater the loss
- 4 of the 10 worst products are Furniture Tables — confirming a structural category problem

3.2 Category & Sub-Category Analysis

Category	Revenue	Profit	Margin %	Orders
Technology	$836,154.10	**$145,455.66**	17.40%	1,544
Office Supplies	$719,046.99	$122,490.88	17.04%	3,742
Furniture	$741,999.98	$18,451.25	**2.49%**	1,764

3.2.1 Loss-Making Sub-Categories

Category	Sub-Category	Revenue	Total Profit	Margin %
Furniture	**Tables**	$206,965.68	**($17,725.59)**	-8.56%
Furniture	Bookcases	$114,880.05	**($3,472.56)**	-3.02%
Office Supplies	Supplies	$46,673.52	**($1,188.99)**	-2.55%

3.2.2 The Discount-Profit Relationship in Furniture

Analysis of all Furniture sub-categories reveals a clear and consistent pattern: every sub-category with an average discount above 20% is loss-making.

Sub-Category	Avg Discount	Margin %	Verdict
Furnishings	14%	+14.24%	**PROFITABLE**
Chairs	17%	+8.10%	**PROFITABLE**
Bookcases	21%	-3.02%	**LOSS-MAKING**
Tables	26%	-8.56%	**LOSS-MAKING**

INSIGHT	The tipping point is clear: once Furniture discounts exceed approximately 20%, the sub-category becomes loss-making. Furnishings at 14% discount achieves 14.24% margin while Tables at 26% discount loses 8.56%. This is a direct and actionable finding.

3.2.3 High Revenue / Low Profit Warning — Machines Sub-Category

- Technology Machines generated $189,238 in revenue but only $3,384 in profit — a margin of just 1.79%
- Average discount on Machines is 31% — the second highest in the entire dataset
- By contrast, Technology Accessories carries only 8% average discount and achieves a 25.05% margin
- Same category, completely opposite profitability — purely driven by discounting decisions

3.3 Regional & State Analysis

Region	Revenue	Profit	Margin %	Orders	Avg Disc.
West	$725,457.93	$108,418.79	14.94%	1,611	11%
East	$678,781.36	$91,522.84	13.48%	1,401	15%
South	$391,721.90	$46,749.71	11.93%	822	15%
Central	$501,239.88	$39,706.45	**7.92%**	1,175	**24%**

3.3.1 Loss-Making States — A Critical Finding

10 out of 49 states are operating at a total loss, collectively destroying $97,687 in profit — equivalent to 34% of the company's total profit.

State	Region	Revenue	Total Loss	Margin %	Avg Disc.
Texas	Central	$170,187.98	**($25,729.29)**	-15.12%	37%
Ohio	East	$78,258.21	($16,971.37)	-21.69%	32%
Pennsylvania	East	$116,512.02	($15,560.04)	-13.35%	33%
Illinois	Central	$80,166.16	($12,607.89)	-15.73%	39%
North Carolina	South	$55,603.09	($7,490.81)	-13.47%	28%
Colorado	West	$32,108.12	($6,527.86)	-20.33%	32%
Tennessee	South	$30,661.92	($5,341.66)	-17.42%	29%
Arizona	West	$35,282.02	($3,427.87)	-9.72%	30%
Florida	South	$89,473.73	($3,399.25)	-3.80%	30%
Oregon	West	$17,431.14	($1,190.48)	-6.83%	29%

CRITICAL	Not a single loss-making state has an average discount below 28%. Every loss-making state in the business carries discounts of 28% or above. This is definitive proof that excessive discounting — not market conditions — is the root cause of state-level losses.

3.4 Customer Segment Analysis

Segment	Revenue	Profit	Margin %	Orders	Avg Discount
Consumer	$1,161,401.34	$134,119.33	11.55%	2,586	16%
Corporate	$706,146.44	$91,979.45	13.03%	1,514	16%
Home Office	$429,653.29	$60,299.01	14.03%	909	15%

- Home Office is the most profitable segment per dollar of sales at 14.03% margin despite being the smallest by volume
- Consumer generates the most absolute profit ($134K) but has the weakest margin (11.55%)
- High discount orders (>50%) cost the Consumer segment $44,562, Corporate $19,191, and Home Office $12,806 — a combined $76,559 in losses purely from over-discounting

3.4.1 Best and Worst Segment-Category Combinations

Segment	Category	Margin %	Verdict
Home Office	Office Supplies	**20.84%**	**BEST COMBINATION**
Corporate	Technology	17.92%	Excellent
Consumer	Technology	17.42%	Strong
Consumer	Furniture	**1.79%**	**CRITICAL**
Home Office	Furniture	3.18%	Poor
Corporate	Furniture	3.31%	Poor

KEY INSIGHT	Furniture is unprofitable across ALL three customer segments. This definitively proves the Furniture problem is not segment-specific — it is a structural pricing and discounting issue with the entire category.

3.5 Shipping & Operational Analysis

Ship Mode	Orders	Avg Days	Revenue	Profit	Margin %
Same Day	264	0	$128,363.12	$15,891.90	12.38%
First Class	787	2	$351,428.43	$48,969.95	13.93%
Second Class	964	3	$459,193.44	$57,446.49	12.51%
Standard Class	2,994	5	$1,358,216.08	$164,089.45	12.08%

3.5.1 The Definitive Discount Band Analysis

This single analysis provides the most actionable finding in the entire report. Grouping all 9,994 orders by discount level reveals the precise financial impact of the Superstore's discounting strategy.

Discount Band	Orders	Avg Disc.	Total Profit	Avg Profit/Order	Avg Margin %
No Discount (0%)	4,798	0%	**$320,987.88**	$66.90	**34.02%**
Low Discount (1-20%)	146	11.8%	$10,448.19	$71.56	11.25%
Medium Discount (21-50%)	4,194	22.1%	$31,520.95	**$7.52**	12.61%
High Discount (51%+)	856	71.9%	**($76,559.23)**	**($89.44)**	**-113.88%**

- No discount orders achieve a 34.02% profit margin — nearly 3x the company average
- Medium discount orders collapse profitability to just $7.52 average profit per order
- High discount orders generate an average loss of $89.44 per order
- Eliminating all high discount orders would add $76,559 to profit — a 26.7% improvement with zero other changes

4. Power BI Dashboard

A three-page interactive dashboard was built in Power BI Desktop, connected directly to the SQL Server database.

Page	Visuals Included	Business Purpose
Page 1: Executive Summary	5 KPI Cards (Revenue, Profit, Loss, Orders, Margin %), Line Chart (Monthly Trend), Bar Chart (Category Performance), Year & Region Slicers	Give leadership an instant snapshot of overall business health and performance trends
Page 2: Product & Category Analysis	Treemap (Category Breakdown), Top 10 Profitable Products, Bottom 10 Products (red), Scatter Plot (Sales vs Profit), Sub-Category Profitability Table with conditional formatting	Identify star products to promote and loss-making products to reprice or discontinue
Page 3: Region, Segment & Customer View	Top/Bottom 10 States Bar Charts, Donut Chart (Segment Split), Profit Margin by Region, Average Discount by Segment, Loss-Making States Table	Surface geographic and segment-level performance gaps to guide regional strategy and marketing investment

5. Business Insights & Recommendations

Based on the full analysis, five critical business insights have been identified. Each is accompanied by a specific, data-backed recommendation.

Insight 1: The Discounting Crisis is the Single Biggest Threat to Profitability

The data tells a clear and consistent story: every major profitability problem in this business traces back to uncontrolled discounting. High discount orders (>50%) are losing $89.44 per order on average. Every loss-making state carries a minimum 28% average discount. The Central region gives 24% average discounts and has the worst margin. Tables carry 26% average discounts and lose $17,726.

RECOMMENDATION	Implement a company-wide discount cap of 20% maximum, with a strict approval process required for any exception above 20%. Based on the data, no business case exists for discounts above 50% — these orders consistently generate losses without exception. Enforcing this policy alone would recover approximately $76,559 in annual profit.

Insight 2: Furniture Must Be Urgently Restructured

Furniture generates $742,000 in revenue — the second highest category — but only 2.49% profit margin. This means the Superstore earns just $18,451 on $742,000 of Furniture sales. Tables and Bookcases are actively loss-making across all customer segments. The problem is entirely discount-driven: Furnishings at 14% discount achieves 14.24% margin while Tables at 26% discount loses 8.56%.

RECOMMENDATION	Immediately cap Furniture discounts at 15% maximum. For Tables specifically, either eliminate discounts entirely or conduct a pricing review to increase base prices. If Tables and Bookcases cannot be made profitable within two quarters under the new discount policy, the Superstore should consider discontinuing these product lines and reallocating shelf and warehouse space to higher-margin categories.

Insight 3: The Central Region Needs Immediate Intervention

The Central region is the clearest example of discount-driven profit destruction. Despite generating $501,000 in revenue, it returns only 7.92% margin — nearly half the West region's 14.94%. Texas alone loses $25,729 at a 37% average discount. Illinois loses $12,608 at 39% average discount. Central accounts for 51% of all high-discount orders in the entire company.

RECOMMENDATION	Launch a Central region discount audit immediately. Identify which sales representatives, product lines, and customer accounts are driving the excessive discounting. Set the Central region maximum discount to 15% — matching the East and South regions. If Central matched the West's discount rate, the potential profit improvement is estimated at $25,000-$30,000 annually.

Insight 4: Invest in Home Office + Office Supplies as the Growth Engine

Home Office customers buying Office Supplies is the single most profitable customer-category combination in the business at 20.84% margin. Home Office is also the segment with the highest average order value ($240.97) and lowest average discount (15%). Yet Home Office has only 909 orders compared to Consumer's 2,586 — it is a significantly underpenetrated market.

RECOMMENDATION	Develop a targeted B2B marketing programme aimed at small home-based businesses and freelancers. Focus promotional activity on Office Supplies bundles for Home Office customers. This segment demonstrates the best unit economics in the business — growing its order volume by even 20% would add approximately $12,000 in incremental profit at current margins.

Insight 5: California and New York are Carrying the Business

California ($76,382 profit) and New York ($74,039 profit) together generate $150,421 — representing 52.5% of the entire company's profit from just 2 states. This extreme concentration is a significant business risk. Meanwhile Florida ($89,474 revenue, -3.80% margin) is close to breakeven and could be flipped profitable with targeted discount reduction. The South region is also underpenetrated with only 822 orders despite decent margins.

RECOMMENDATION	Protect and invest in California and New York — these states must not be disrupted by price changes or inventory reductions. Simultaneously, prioritise Florida as the highest-potential recovery state (close to breakeven, high revenue base). Develop a South region growth strategy targeting Texas and North Carolina with restructured pricing, not higher discounts.

6. Conclusion

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This project demonstrates how data analytics can be used to uncover hidden profitability challenges within a growing retail business. Although revenue increased by 51% between 2014 and 2017, profitability did not improve at the same pace due to the widespread use of excessive discounts.

Across products, categories, customer segments, regions, and states, a consistent pattern emerged: higher discounts were strongly associated with lower profitability and, in many cases, outright losses. The analysis identified discount management as the most significant opportunity for improving financial performance.

Key recommendations include implementing discount controls, restructuring pricing within the Furniture category, improving profitability within the Central region, expanding investment in high-margin customer segments, and strengthening performance in growth markets.

From a business perspective, this project highlights the importance of balancing revenue growth with profitability. From an analytics perspective, it demonstrates how data-driven insights can support strategic decision-making and operational improvement.

Key Takeaway

Revenue growth alone does not guarantee business success. Sustainable growth requires disciplined pricing strategies, effective cost management, and continuous monitoring of profitability drivers.